

COLLARDRIVE · DESIGN DECISION RECORD

CollarDrive design for construction

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DOCUMENT CONTROL

Document ID	CLD-DDR-002	Version	0.1
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/collardrive		
Source	docs/decisions/0002-design-for-construction.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	Changes that make the concept constructable, decided under Amish's 2026-10-03 pre-approval

0002: Design for construction

- **Date:** 2026-10-03
- **Status:** accepted. Decided by Amish under his pre-approval of 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost." None of the changes alters what CollarDrive does, its pitch or its safety case, except to make the safety case stronger (backstop, guards, sleepers set back from the collar).

■ Context

The TRL 2 concept was a massing model: one welded frame "bolted and staked over the collar", an engine, a layshaft, a wheel with "a clutch", one belt to the pump, a blower, a duct and an exhaust extension. Writing the build plan (CLD-BLD-001) meant deciding how every part is made and how it joins its neighbours, following Amish's rule of 2026-09-30: "fix the design assumptions to match and be physically feasible". The model `cad/src/model.py` was changed until its fit check (`python cad/src/model.py --check`) reports no two components overlapping and every joint face touching (48 components, 0 overlaps, 0 missing contacts).

Safety: These changes concern a machine with a running engine, belts and a rope beside an open shaft. The backstop, the guards and the set-back sleepers are safety features and are not to be left out of a build. The frame is not a working platform; the collar is fenced; the gas detector is used below ground.

■ Options considered

For each problem the simplest physically sound fix that keeps what the concept does was chosen. Alternatives considered are named in the "Why" column of Table 1.

■ Decision

Table 1. Changes made for construction.

#	Component	The concept had	The constructable design has	Why
1	Frame	One welded frame about 3.9 m long, over 100 kg	Two modules bolted at a joint: a span module 2.75 m long (69.6 kg with its pedestals and cleats) and a drive module 1.15 m long (34.3 kg); 10 mm joint plates and eight M16 grade 8.8 bolts	Each piece can be carried by hand to a remote collar; a single frame would need lifting gear
2	Ground support	"Bolted and staked over the collar", bearing on the collar	Three 200 x 100 x 900 mm timber sleepers, their edges at least 0.35 m back from the edge of a 1.5 m collar, each held by two 25 mm stakes 700 mm into the ground; welded cleats and M12 coach screws hold the sills to the sleepers	Nothing loads the collar lip, which may crumble; the stakes stop the frame walking under belt pull
3	Pump belt	One belt from the layshaft to the wheel	Two matched A belts on a 90 mm two-groove layshaft pulley and a 560 mm two-groove cast pump pulley	One belt would carry about 184 % of its allowable pull at 30 m; two carry 92 %
4	Clutch	"Clutch" on the wheel, not defined	Over-centre idler on the slack strand of the pump belts, a lever through the belt guard with a latch plate	No bought clutch; the blower runs alone with the lever down
5	Bearings	Shafts floating at their heights	UCP206 and UCP205 pillow blocks on welded pedestals of SHS 60 x 60 x 4 with 10 mm top plates	The shafts must sit at set heights above the sills
6	Backstop	None	12-tooth ratchet keyed to the pump shaft and a sprung pawl on a post welded to a crossbar	A 30 m water column runs the rope back with 61 N·m when the engine stops
7	Pump wheel	A disc	Steel hub, 6 mm web and two car tyre sidewalls bolted to form a V that grips the rope on a 400 mm circle	The open rope pump wheel, buildable anywhere
8	Pipe head	Pipe "hung down the shaft"	8 mm plate across two crossbars with a split collar clamping the main; outlet tee with a rope exit stub above it; spout to the front; M12 eye bolt for a 6 mm support wire	The main and the guide block hang from the wire and the plate, not from the PVC
9	Guide block	Not drawn	Weighted 6 mm steel box with an HDPE roller on a stainless pin, an inlet socket and slotted sides	Turns the rope into the main at the bottom and keeps it down
10	Duct head	Duct "from the blower"	Rigid 200 mm galvanised bend on a saddle and strap on a crossbar; a 0.7 m flexible link to the blower outlet	The lay-flat duct needs a rigid turn and a fixed point
11	Engine and blower mounting	Sitting on the frame	8 mm plates with four slots each, bolted to crossbars placed under them; engine and blower bolted to the plates	Slots tension the belts; crossbars carry the feet
12	Guards	"Belt and pulley guards"	Belt guard of angle, sheet and 12.7 mm mesh on three brackets; wheel hood; layshaft cover; 6 mm mesh blower inlet guard	Every moving part enclosed (R9)
13	Exhaust			

#	Component	The concept had	The constructable design has	Why
		"Extension carries fumes downwind"	Stainless flexible hose, 40 mm galvanised pipe on two low stands, bend and riser on a tall stand, outlet 2.4 m high with a rain cap, 7.8 m from the shaft centre	Meets the 6 m distance with the riser above head height
14	Water outlet	Not drawn	Spout to the front of the frame and a 10 m discharge hose	Water laid at least 5 m from the collar
15	Crossbars	Not placed	Six crossbars in each module, placed under the pedestals, pawl post, pipe head plate, duct saddle and the two plates	Every fixing has steel under it

■ Consequences

- The model, STEP and STL exports, the general arrangement CLD-DWG-001 (Rev P2) and the concept media were regenerated from the changed model.
- The calculations were rerun (CLD-CAL-001 v0.2); every requirement is met on paper.
- The bill of materials gained lines for sleepers and stakes, joint bolts, pedestals, the backstop, the clutch, the pipe head, the guide block, the support wire, the duct head and the discharge hose. Estimated cost USD 2,535 against the USD 3,500 value-engineering target.
- Facts that can only be settled with real parts (pillow block bolt centres, blower feet, engine base, belt lengths) are listed under "To confirm when parts are bought" in the design decisions register (CLD-DEC-001).